Six math clues · answer key

How Holograms Work

Add or subtract the ones first; trade a ten when the ones need it.

Math 1 · 25

62 − 37 = 25. Trade a ten for ones, then take away the squares with a bird.

Math 2 · 63

27 + 36 = 63. 7 + 6 makes 13, so trade 10 ones for a ten.

Math 3 · 26

54 − 28 = 26. Trade a ten for ones, then take away the dark stripes.

Math 4 · 43

90 − 47 = 43. Trade a ten for ones, then take away the first piece.

Math 5 · 45

19 + 26 = 45. 9 + 6 makes 15, so trade 10 ones for a ten.

Math 6 · 83

45 + 38 = 83. 5 + 8 makes 13, so trade 10 ones for a ten.

Three additions and three subtractions, each with regrouping. The numbers are for practice, not real hologram counts.

Reading & writing · answer key

How Holograms Work

Reading · 1

Tilting shows different sides of the bird, so it seems to sit inside the card.

Reading · 2

A bump and a dip cancel into a calm line.

Reading · 3

The film records stripes where two laser beams crossed, not a picture.

Reading · 4

Light from every spot on the shell reached every part of the film.

Reading · 5

Grooves bend light back into the bird's waves; each eye sees a different view.

Reading · 6

A reflection in slanted clear glass (Pepper's ghost).

Math · all

25, 63, 26, 43, 45, 83.

Screen-free

They cancel into calm; fine stripes; the whole shell; a reflection in slanted glass.

Lasers are not toys: never look into one or point one at eyes. Use only a little water in the pan and wipe up spills. Keep glass away from little hands; a clear plastic sheet works for the window trick.

Offer dictation: let the child say a complete thought while you write it, then invite a picture or label.

Four looks at a hologram

How Holograms Work

Quiz 1. What happens where a bump meets a dip?

A bump and a dip cancel, so the water goes calm. Light waves can do this too.

Quiz 2. What is on a hologram's film?

The film records the stripes where two laser beams crossed.

Quiz 3. What does half a hologram show?

Every part of the film saw the whole shell, so each half shows all of it.

Quiz 4. How do most floating stage pictures work?

Slanted clear glass reflects a lit figure, so it seems to float.

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